Accepted Manuscript

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PII: S0269-7491(18)30787-5

DOI: 10.1016/j.envpol.2018.08.088

Reference: ENPO 11526

To appear in: *Environmental Pollution*

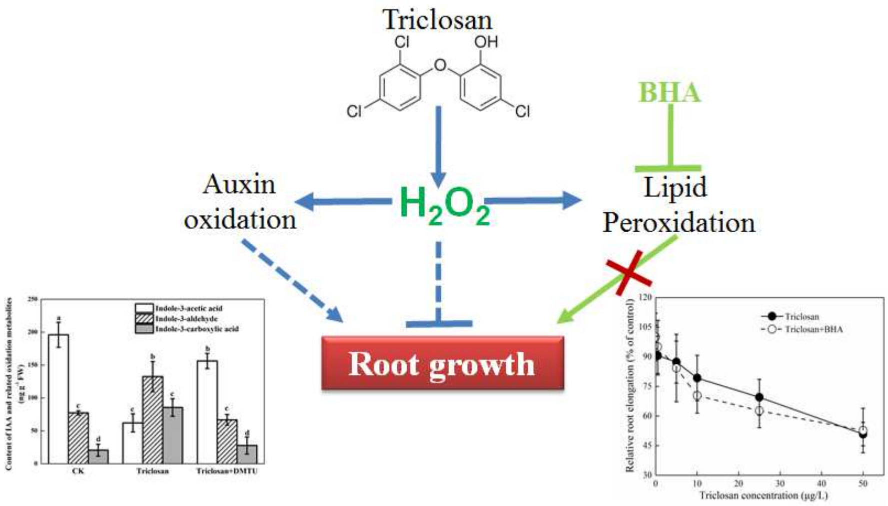
Received Date: 23 February 2018

Revised Date: 3 August 2018

Accepted Date: 26 August 2018

Please cite this article as: Sun, C., Dudley, S., McGinnis, M., Gan, J., Hydrogen peroxide mediates triclosan-induced inhibition of root growth in wheat seedlings, *Environmental Pollution* (2018), doi: 10.1016/j.envpol.2018.08.088.

Graphical abstract



Title: Hydrogen peroxide mediates triclosan-induced inhibition of root growth in wheat seedlings

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Abstract

Triclosan, an extensively used antimicrobial agent, enters agroecosystems when sewage sludge and reclaimed water are applied to agricultural fields, and may trigger a series of plant physiological and biochemical responses. However, few studies have investigated the mechanism by which plant development is affected by triclosan. Here, microscopic, pharmacological and biochemical analyses, and histochemical dye staining were used to explore the effects of triclosan on root growth in wheat plants. Exposure to triclosan inhibited root elongation, and significantly triggered hydrogen peroxide (H_2O_2) production and lipid peroxidation in wheat roots. The inhibition of root growth by triclosan was reversed by dimethylthiourea, a H_2O_2 scavenger, indicating that alterations of endogenous H_2O_2 concentrations in root cells were likely linked to triclosan-induced root growth inhibition. The addition of butylated hydroxyanisole, a lipophilic antioxidant, during triclosan treatment completely prevented the increase of lipid peroxidation, but did not alleviate triclosan-induced reduction of root growth. In triclosan-treated wheat roots, the level of indole-3-acetic acid decreased by 68.3%, while the contents of two indole-3-acetic acid oxidative metabolites, indole-3-aldehyde and indole-3-carboxylic acid, increased by 71.3% and 314.4%, respectively. Moreover, the oxidation of auxin induced by triclosan in wheat roots was prevented by dimethylthiourea. These results together suggested that the triclosan-enhanced production of H_2O_2 induced auxin oxidation, thus leading to the suppression of root growth. Findings of this study improve our mechanistic understanding on how antimicrobial agents such as triclosan affect plant root growth.

Key words: triclosan; root growth; lipid peroxidation; hydrogen peroxide; emerging contaminants.

Introduction

Triclosan, a broad-spectrum bacteriostat and fungicide, is widely used in personal care products, cleaning agents, athletic clothing and food packaging (Yueh and Tukey, 2016). Approximately 96% of triclosan in consumer products is washed down the drain, leading to its aggregation at wastewater treatment plants (WWTPs) (McAvoy *et al.*, 2002). Owing to its incomplete removal at WWTPs, triclosan ranks among the priority contaminants of concern worldwide (Halden, 2014; von der Ohe *et al.*, 2012).

Due to its intermediate hydrophobicity, triclosan appears in both WWTP effluent and biosolids (Heidler and Halden, 2007; Yueh and Tukey, 2016). When biosolids and treated wastewater are used on agricultural fields, triclosan enters agroecosystems, posing unintended risks, including phytotoxicity. In particular, plant roots are the first point of contact for such emerging contaminants, and play pivotal roles in the inception and manifestation of phytotoxicity. Earlier studies showed that triclosan was primarily found in the roots (Corrotea *et al.*, 2016) and was capable of inhibiting root growth of certain plants, including wheat, rice and cucumber (An *et al.*, 2009; Liu *et al.*, 2009). However, research to date has yet to reveal the corresponding physiological and molecular mechanisms, including changes in cellular metabolism and signaling, that underline such inhibitions.

Accumulating evidence suggests that stress-induced H_2O_2 plays an important role in the regulation of root system development (Liszkay *et al.*, 2004, Dunand *et al.*, 2007; Tsukagoshi, 2016). Ahammed *et al.* (2017) suggested that the reduction of root growth in *Cucumis sativus* by organic pollutants, like 2,4,6-trichlorophenol, chlorpyrifos and oxytetracycline, was caused by an over-accumulation of reactive oxygen species; while Yu *et al.* (2017) observed that inhibition of H_2O_2 accumulation in the root meristem contributed to oxytetracycline-induced suppression of root growth in tomato plants. These results suggested that alterations of endogenous H_2O_2 levels in root cells were likely linked to xenobiotic-induced growth inhibition; however, the corresponding physiological roles and molecular mechanisms of H_2O_2 in inhibiting root elongation under stress by organic pollutants remain elusive.

Auxin is a key regulator of root growth and is sensitive to environmental stimulus (Teale *et al.*, 2006), including pharmaceutical and personal care products (PPCPs). For instance, carbamazepine and verapamil were capable of disrupting auxin homeostasis (Carter et al., 2015). There is growing interest but limited research in H_2O_2-auxin interactions in regulating the growth and development of plant roots under various xenobiotics (Jansen *et al.*,2001; Zhao *et al.*, 2012). For example, H_2O_2 affected auxin stability by inducing IAA degradation in roots of tobacco (Jansen *et al.*,2001). In a recent study, H_2O_2 induced by fluazifop-p-butyl, a herbicide, disturbed auxin signaling in shoots of *Acanthospermum hispidum* (Liu *et al.*, 2017). However, questions remain as to how actions of auxin and H_2O_2 are coupled in influencing root growth and development under PPCP stress.

It may be hypothesized that H_2O_2 over-accumulation disrupts auxin signaling in mediating the regulation of root development after triclosan exposure. The combined use of physiological and pharmacological approaches with specific inhibitors allows the elucidation of plant-xenobiotic interactions. Furthermore, the complementary use of historical approaches facilitates the detection of oxidative events in single cells of a tissue composed of different cell populations. The objectives of this study were to investigate the role of H_2O_2 in triclosan-induced inhibition of root growth in wheat seedlings, and to examine the interactions between H_2O_2 and auxin in regulating root elongation after triclosan exposure.

Materials and Methods

Plant materials and treatment

Analytical standard of triclosan (purity >98%) was purchased from Alfa Aesar (Ward Hill, MA). Triclosan-d_3 was obtained from C/D/N Isotopes (Pointe-Claire, Quebec, Canada). Stock solutions of triclosan at 5, 50, 100, 250 and 500 mg L^{-1} were prepared in methanol and stored at -20 °C before use. Indole-3-acetic acid, indole-3-aldehyde, and indole-3-carboxylic acid were acquired from Santa Cruz Biotechnology (Santa Cruz, CA). Solvents, including acetonitrile, methanol, and formic acid, were Ultima grade (Fisher Scientific, Fair Lawn, NJ). All other chemicals and reagents used in biochemical measurements were of analytical grade or better. Purified water was prepared using a Milli-Q system (Millipore, Carrigtwohill, Cork, Ireland).

Seeds of wheat were sterilized with 1% (v/v) NaClO for 20 min, rinsed with deionized water three times, and then germinated in the dark before cultivation in 2.5 L of Hoagland nutrient solution. Germination and growth occurred in a growth chamber under a 12 : 12 h, 25 : 22 °C day: night regime at a light intensity of 300 μmol m^{-2} s^{-1} and a relative humidity of 70%. The solution was renewed daily. After 3 d of cultivation, uniform seedlings were transferred to 1.25 L of Hoagland nutrient solution that was spiked with 100 μL of the triclosan stock solution, and cultivated for 24 h. To assess the effects of various inhibitors and scavengers, 3-d old seedlings were treated with triclosan in the presence of butylated hydroxyanisole (BHA, 10 μM, Acros) or dimethylthiourea (DMTU, 2.5 mM, Sigma Aldrich) for 24 h. The concentrations used in this study were based on preliminary experiments, and BHA (dissolved in methanol) or DMTU (dissolved in water) was added to the solution just prior to the addition of triclosan. Each treatment was repeated at least three times.

Relative root elongation

Seedlings with a root length about 5 cm were incubated in control solution (full-strength Hoagland nutrient solution, pH = 5.5) or treatment solutions containing 0.5, 5, 10, 25 and 50 μg L^{-1} triclosan as well as 25 μg L^{-1} triclosan with 10 μM BHA or 2.5 mM DMTU for 24 h. Elongation of the root was measured with a ruler before and after the treatment. Relative root elongation was calculated as the percentage of the root elongated by triclosan treatment as compared to the control. Relative root elongation was calculated according to the following equations:

$$\text{Relative root elongation (\%)} = \frac{RE_{\mathrm{t}}}{RE_{\mathrm{c}}} \times 100$$

where RE_t and RE_c are the mean root elongation by triclosan and mean root elongation by control, respectively.

Lipid peroxidation measurement

The level of membrane lipid peroxidation was estimated in terms of malondialdehyde as described in Wang *et al.* (2010). Detailes of extraction and determination for malondialdehyde are provided in Text S1 in the Supporting Information.

H_2O_2 determination

The content of H_2O_2 in the wheat root tissues was determined spectrophotometrically at 415 nm by monitoring the titanium-peroxide complex as in Sun *et al.* (2014). Fresh roots (0.2 g) were frozen in liquid nitrogen and homogenized in 2.5 mL cold acetone on ice. The homogenate was centrifuged at 5000 g for 10 min at 4 °C. A 1.0 mL aliquot of the supernatant (blank control) was added catalase (100 Units/mL) and then kept at the room temperature for 10 min. Another 1.0 mL aliquot of supernatant as well as the blank control were mixed with 0.1 mL of 5% $TiSO_4$ and 0.1 mL ammonia. After centrifugation, the titanium-peroxide complex pellet was resuspended in 3.0 mL of 2 M H_2SO_4, and then the absorbance was measured on a Cary 50 UV-Visible spectrophotometer. The H_2O_2 content was calculated using an external standard curve generated with known concentrations of H_2O_2 (0-1 μM, r^2=0.98). The concentration of H_2O_2 in plant samples was calculated by subtracting the catalase insensitive background from the measured value.

Histochemical analyses

Lipid peroxidation was visualized by Schiff's reagent (Sigma-Aldrich) and the loss of plasma membrane integrity was detected by staining roots with an Evans blue solution (0.025%, w/v), as described in Yamamoto *et al.* (2001). H_2O_2 was stained by

3,3-diaminobenzidine (DAB, Sigma-Aldrich) as substrate, according to Zhang *et al.* (2007) (Text S2). All stained roots were photographed immediately.

Determination of IAA and two oxidative metabolites

Contents of indole-3-acetic acid (IAA) and its two oxidative metabolites (indole-3-aldehyde and indole-3-carboxylic acid) were determined according to Liu *et al.* (2017) with minor modifications. Root apexes (0-20 mm) was thoroughly homogenized in liquid nitrogen and suspended in 5 mL cold methanol overnight at -20 °C. The homogenate was then centrifuged at 8000 g for 10 min at 4 °C. The residues were further extracted twice with cold methanol and the extracts were combined. The pooled supernatant was concentrated to near dryness under nitrogen, and was then re-dissolved in 1 mL methanol, followed by addition of 20 mL deionized water. The extract was loaded onto a C18 cartridge preconditioned with 3 mL deionized water, followed with 3 mL methanol. The cartridge was eluted with 2 mL methanol under gravity, and the eluate was filtered (PTFE, 0.2 mm, Millipore, Carrigtwohill, Cork, Ireland) before analysis.

Samples were analyzed using a Waters ACQUITY ultra-performance liquid chromatography (UPLC) in combination with a Waters Micromass Triple Quadrupole mass spectrometer (TQD) equipped with an electrospray ionization (ESI) interface (Waters, Milford, MA). The details of instrument analysis are provided in Table S1 in the Supporting Information.

Measurement of triclosan tissue concentrations

A 0.8-g aliquot of plant roots (fresh weight) was extracted with methyl tert-butyl ether (MTBE) and acetonitrile according to Fu *et al.* (2016a). Then the extract was loaded onto an Oasis™ HLB cartridge (150 mg, Waters, Milford, MA) that was preconditioned with 6 mL methanol and 12 mL deionized water, and eluted with 15 mL methanol under gravity. The cleaned eluate was evaporated to dryness under nitrogen, and the residue was recovered in 1.5 mL methanol:water mixture (v:v; 1:1). The final extract was filtered through a 0.2 μm PTFE membrane (Millipore, Carrigtwohill, Cork, Ireland) before analysis.

Triclosan was quantified on the Waters UPLC-QqQ system using a BEH C18 column at a flow rate of 0.3 mL min^{-1}. Mobile phase A was water (containing 0.002% formic acid) and mobile phase B was methanol. Triclosan and the internal standard triclosan-d_3 were detected in the negative ESI mode. The details of instrument analysis are provided in Table S2 in the Supporting Information.

Quality control and statistical analysis

In situ histochemical staining for lipid peroxidation and H_2O_2 were repeated at least 10 times. The contents of H_2O_2, auxin and oxidative metabolites were immediately extracted on ice and analyzed in the dark after sampling, and each was performed in triplicate. Data were statistically analyzed using the SPSS package (version 11.0; SPSS, Chicago, IL, USA). The mean and standard deviation (SD) of each treatment were calculated. Statistical differences in H_2O_2 and lipid peroxidation between treated and untreated plants (at the same sampling point) were performed by the analysis of variance (one-way ANOVA) with Least Significant Difference (LSD) post-hoc analysis ($p < 0.05$ and $p < 0.01$).

Results and discussion

H_2O_2 production and oxidative damage caused by triclosan in wheat roots

To assess the oxidative stress induced by triclosan exposure in wheat roots, H_2O_2 production, lipid peroxidation and the loss of plasma membrane integrity were detected by histochemical staining with DAB solution, Schiff's reagent and Evans blue, respectively, after exposure to 25 μg L^{-1} triclosan for 24 h. *In situ* staining showed clearly that triclosan treatment significantly stimulated H_2O_2 production (Fig. 1a), consequently caused lipid peroxidation (Fig. 1b) and further compromised plasma membrane integrity (Fig. 1c). The mechanistic basis of the process is presumably through triclosan-induced mitochondrial dysfunctions (Ajao *et al.*, 2015), leading to the inhibition of normal oxygen consumption (i.e., respiration inhibition) and enhancement of ROS production. These results, similar to those reported by An *et al.* (2009) on wheat roots, suggested that oxidative stress may be a common symptom in plant response to triclosan.

Spatio-temporal distribution of H_2O_2 and lipid peroxidation

Inhibition of root elongation increased with increasing external triclosan concentrations (Fig. 2a). Even at 0.5 μg L^{-1}, triclosan inhibited root elongation by 10.5%, and the inhibition further increased to 30.5% when the level of triclosan was 25 μg L^{-1}. A similar behavior was reported earlier in three wetland plants, i.e., *Sesbania herbacea*, *Eclipta prostrata*, and *Bidens frondosa* (Stevens *et al.*, 2009), and two terrestrial crops, i.e., cucumber and rice (Liu *et al.*, 2009). It must be noted that the exposure concentrations used in this study fell within the range of environmentally relevant concentrations. In an actual soil-plant system, plant roots are exposed to contaminants such as triclosan primarily in the soil porewater. Assuming a K_{oc} value of 1.1×10^4 L kg^{-1} in a soil containing 1.5% organic carbon for triclosan (Fu *et al.*, 2016b), a bulk soil concentration of 1 mg kg^{-1} would translate to 6.0 μg L^{-1} for the soil porewater at 10% soil water content. To test whether the inhibition of root elongation by triclosan would also occur in realistic field environments, wheat plants were cultivated in soil spiked with or without 10 mg kg^{-1} triclosan (Fig. S1a). It was observed that root elongation was significantly ($P < 0.01$) inhibited (Fig. S1a,b). Liu *et al.*(2009) also reported that triclosan inhibited root elongation of rice and cucumber plants grown in soil. Our results, if coupled with previously published studies, reinforced the possibility that contamination of agricultural soils by triclosan from reclaimed water irrigation or biosolids amendment may cause pronounced phytotoxicity to agricultural crops.

Over the wide range of triclosan concentrations, triclosan accumulation (Fig. 2b) in the root was negatively correlated with root elongation, and the correlation was statistically significant (r^2=0.92; $p < 0.01$). In addition to inhibit root elongation, triclosan also strongly induced H_2O_2 production (Fig. 2c) and lipid peroxidation (Fig. 2d). The H_2O_2 content in roots without triclosan treatment were relatively constant along the root length, whereas H_2O_2 content in the treated roots was approximately 2-fold that in the control and decreased from the root tip toward the base (Table 1).

Although lipid peroxidation in the triclosan treated roots was similar to the control in sections of the basal region (3-4 and 4-5 cm), an enhancement of lipid peroxidation was evident in the 0-3 cm region (Table 1). However, after 25 μg L^{-1} triclosan treatment, the highest triclosan level was observed in the 1-2 cm from the root tip toward the basal region (Table 1). This was different from metal stress, where root tips were usually found to be the main site for metal accumulation (Sun *et al*., 2015; Yamamoto *et al*., 2001).

Time-dependent concentrations of triclosan in wheat roots are depicted in Fig. 3a. The level of triclosan reached the maxima at 9 h in the roots, followed by a rapid decrease thereafter. This was likely due to the translocation of triclosan to the aboveground parts and/or active biotransformation in the roots (Macherius *et al.*, 2012). In a previous study, the translocation factor of triclosan was estimated to be 0.001-0.1 in common vegetables including lettuce, spinach, cucumber, and pepper (Wu *et al.*, 2013), suggesting limited translocation once triclosan is taken up. Therefore, the rapid dissipation of triclosan from the wheat root tissues may be attributed mainly to *in situ* metabolism catalyzed by a cascade of enzymes. For example, incubation of triclosan in carrot cell cultures or in intact plants showed that triclosan was converted to multiple conjugated metabolites in plant cells. Other transformation products included triclosan conjugates with saccharides, disaccharides, malonic acid, and sulfate (Macherius *et al.*, 2012). On the other hand, the kinetic patterns of H_2O_2 production (Fig. 3b) and lipid peroxidation (Fig. 3c) were similar to each other, with the oxidative damage occurring at 6 h and exacerbated thereafter.

Triclosan-induced inhibition of root growth became more pronounced over time; the average root length was reduced from 19.3 ± 2.3 mm to 12.7 ± 2.6 mm after 24 h of cultivation in the triclosan-treated solution (Fig. 3d). As a whole, these results showed that triclosan, under environmentally relevant conditions, induced H_2O_2 over-accumulation, caused extensive lipid peroxidation and inhibited root elongation.

Cause and effect relationship between H_2O_2 and root inhibition

The role of triclosan-induced H_2O_2 in root growth inhibition was investigated by using DMTU, a H_2O_2 scavenger. Relative root elongation of wheat under treatment of triclosan + DMTU was 97.6%, but only 65.5% for the triclosan treatment alone. These results suggested that H_2O_2 was involved in triclosan-induced root growth inhibition. However, elimination of H_2O_2 by DMTU had no discernable effect on triclosan uptake and accumulation in wheat roots (Fig. 4b). These results implied that the recovery of root elongation following scavenging H_2O_2 was unlikely a result due to decreases in triclosan accumulation in wheat roots by DMTU. Recently, H_2O_2 was demonstrated to play a role in xenobiotic-induced inhibition of root growth of various plants (Chao *et al.*, 2009; Ahammed *et al.*, 2017). It has been suggested that H_2O_2 contributes to increasing lignin formation, enhancing cell wall rigidification, restricting cell expansion and disturbing auxin signaling, leading to the ultimate inhibition of root elongation (Liszkay *et al.*, 2004; Xiong *et al.*, 2015). This finding, however, was different from a previous study showing that oxytetracycline, a widely used antibiotic, induced arrest of root growth was associated with decreased H_2O_2 levels in tomato seedlings (Yu *et al.*, 2017). It is possible that the discrepancy between the studies may be attributed to different xenobiotics. However, these studies nevertheless indicate that disruption of H_2O_2 homeostasis is involved in emerging contaminant-induced inhibition of root elongation.

Application of exogenous BHA (10 μM) to the culture medium efficiently prevented lipid peroxidation, as verified by histochemical staining with Schiff's reagent (Fig. 5a). This was further confirmed by measurement of the content of malondialdehyde, a major product of lipid peroxidation. With the co-treatment of BHA, the content of malondialdehyde in the wheat root exposed to triclosan was 1.06 fold the control, while it was 1.57 fold the control for the triclosan-alone treatment (Fig. 5b). A similar effect of BHA on the peroxidation of lipids under xenobiotic stress was reported in other plants including *Hibiscus moscheutos* and *Pisum sativum* (Tian *et al.*, 2007; Yamamoto *et al.*, 2001). The level of triclosan in wheat roots after BHA application was 549 ng g^{-1} (fresh weight), which was similar to the treatment without BHA application (Fig. 5c). These results indicated that the alleviated lipid peroxidation by BHA was probably not associated with triclosan accumulation in wheat roots. In addition, the pretreatment of BHA at different triclosan exposure rates did not appreciably influence triclosan-induced inhibition of root elongation (Fig. 5d). This observation highlighted that triclosan-induced root inhibition was not directly related to H_2O_2-mediated peroxidation of lipids. In cucumber plants, however, it was found that various organic pollutants, such as 2,4,6-trichlorophenol, chlorpyrifos and oxytetracycline, restricted root growth by inducing H_2O_2 production, disrupting redox homeostasis and increasing lipid peroxidation (Ahammed et al., 2017). The discrepancy among the studies may be attributed to different plant species and chemical properties. Therefore, lipid peroxidation was likely an early symptom triggered by triclosan, but not the primary cause of elongation inhibition in wheat roots; and H_2O_2 likely had other functions in the regulation of root growth under stress by triclosan.

IAA oxidation and triclosan-induced root growth inhibition

In this study, the level of indole-3-acetic acid (IAA) after triclosan treatment decreased significantly as compared to the control in the hydroponic experiments after 24 h, with the content of IAA declining from 196 ng g^{-1} FW to 62 ng g^{-1} FW (Fig. 6). When wheat seedlings were cultivated in soil spiked with 10 mg kg^{-1} triclosan, decreases in IAA were also detected (Fig. S1c). Previous studies showed that H_2O_2 disrupted auxin signaling by affecting the concentration and distribution of auxin and the expression of auxin biosynthesis, polar transport, and responsive genes (Jansen *et al.*, 2001; Chaoui and El Ferjani, 2005). For example, H_2O_2 inhibited root growth by influencing auxin signaling in fluazifop-p-butyl-stressed *Acanthospermum hispidum* (Liu *et al.*, 2017). Considering that IAA is an essential plant hormone for the regulation of root growth, it was likely that triclosan exerted its phytotoxicity by affecting the synthesis and stability of auxin in wheat.

In the present study, the contents of the two main oxidative metabolites of IAA, i.e., indole-3-aldehyde and indole-3-carboxylic acid, were further quantified. In contrast to the control treatment, these two IAA oxidative metabolites increased in the triclosan-treated wheat seedlings by 71.4% and 314.4%, and 87.1% and 201.3% in the hydroponic and soil experiments, respectively (Fig. 6; Fig. S1c). Treatment with a scavenger of H_2O_2 (DMTU) significantly restored IAA levels after triclosan treatment in wheat roots (Fig.6), and further alleviated triclosan-induced root growth inhibition (Fig. 4a). Hydrogen peroxide may affect auxin stability by oxidizing IAA via the induction of the peroxidase activity (Kawano, 2003), as increased peroxidase activity was observed in wheat leaves after exposure to 0.2 mg L^{-1} triclosan (An *et al.*, 2009). These studies suggest that peroxidase was probably involved in IAA oxidation under triclosan stress. It has been suggested that the specificity of plant peroxidase to IAA may be possibly conferred by the auxin-binding domains near the heme pocket (Kawano *et al.*, 2003). In a recent study, IAA oxidation was also found to be involved in the mode of action of fluazifop-p-butyl, a herbicide, in *Acanthospermum hispidum* (Liu *et al.*, 2017). However, it must be noted that results from the current study were insufficient to exclude the possibility that the inhibition of IAA biosynthesis contributed to the decrease in the IAA content in wheat roots during triclosan exposure. Therefore, further research is needed to understand the exact functions of H_2O_2 in auxin signaling, by considering its effects on biosynthesis, distribution and transport of auxin under the stress of triclosan and similar contaminants.

Conclusions

In summary, the present study revealed a negative role of endogenous H_2O_2 production in triclosan-induced root growth inhibition in wheat plants. The mechanistic basis of the process is presumably through disruption of the auxin signaling pathway, including auxin oxidation, thereby altering the growth of the root system, not by H_2O_2-induced lipid peroxidation. Findings from this study provided evidence that triclosan was capable of eliciting phytotoxic effects on agricultural crops such as wheat under environmentally relevant concentrations. However, to better understand the potential risks of emerging contaminants in agroecosystems owing to the increased use of sewage sludge and reclaimed water in agricultural fields, future research must be devoted to elucidating their precise phytotoxicity mechanisms.

Acknowledgements

This work was financially supported by the USDA-National Institute for Food and Agriculture Award No. 2016-67017-24433.

Supporting Information

Additional details on descriptions of lipid peroxidation measurement, histochemical detection of H_2O_2, and instrumental method for the analysis of indole-3-acetic acid, indole-3-aldehyde, indole-3-carboxylic acid, triclosan and triclosan-d_3 by UPLC-MS/MS are available free of charge online.

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459 **Table 1.** Triclosan content, H_2O_2 content and lipid peroxidation along wheat roots

460 caused by 25 μg L^{-1} triclosan. The root apex was sectioned in 1 cm intervals from the

461 tip to towards the basal region (5 cm).

Distance from root tip (cm)	**Triclosan content (ng g^{-1} FW)**		**H_2O_2 content (μmol g^{-1}FW)**		**Lipid peroxidation (nmol TBARS g^{-1} FW)**	
	CK	Triclosan	CK	Triclosan	CK	Triclosan
0-1	Null	799±56	11.8±0.8	21.4±2.1**	3.2±0.5	8.8±0.9**
1-2	Null	1440±42	9.9±1.6	18.1±1.0**	2.3±0.7	6.7±0.4**
2-3	Null	1041±30	11.6±1.9	16.2±1.5**	2.3±0.4	5.1±0.9**
3-4	Null	679±38	10.2±2.1	15.2±1.3**	2.3±0.2	3.1±0.9
4-5	Null	608±97	6.7±0.4	12.0±2.6**	1.9±0.3	2.1±0.1

462 **, Significant difference between triclosan and control treatments at $p < 0.01$.

Figure Legends

Figure 1. Histochemical detection of (a) H_2O_2, (b) lipid peroxidation and (c) plasma membrane integrity in roots of wheat seedlings exposed to triclosan at 25 μg L^{-1}.

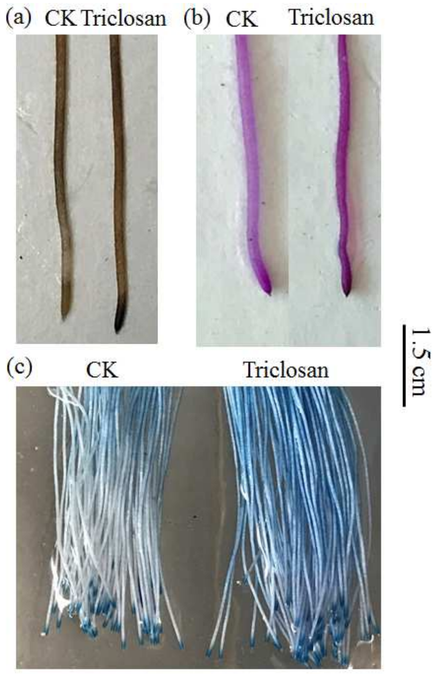
Figure 2. Triclosan dose dependency of (a) root elongation, (b) triclosan accumulation, (c) H_2O_2 content and (d) lipid peroxidation in wheat roots.

Figure 3. Time course of (a) triclosan accumulation, (b) H_2O_2 content, (c) lipid peroxidation and (d) root elongation of wheat seedlings exposed to triclosan at 25 μg L^{-1}. **, significant difference between triclosan and control treatments at $p < 0.01$.

Figure 4. Effect of H_2O_2 scavenger (dimethylthiourea, DMTU) on (a) root elongation and (b) triclosan accumulation of wheat seedlings exposed to triclosan at 25 μg L^{-1}. Different letters indicate significant difference ($p < 0.05$) between the treatments.

Figure 5. Effect of lipophilic antioxidant (butylated hydroxyanisole, BHA) on lipid peroxidation, triclosan accumulation and root elongation. (a) Histochemical detection of lipid peroxidation; (b) Biochemical detection of lipid peroxidation; (c) Triclosan accumulation in wheat roots; and (d) Effect of BHA on the triclosan-induced inhibition of root elongation in wheat roots. *, Significant difference ($p < 0.05$) between the treatments.

Figure 6. Effects of triclosan and H_2O_2 scavenger (dimethylthiourea, DMTU) on levels of indole-3-acetic acid (IAA) and its two oxidative metabolites (indole-3-aldehyde and indole-3-carboxyloic acid) in wheat roots.



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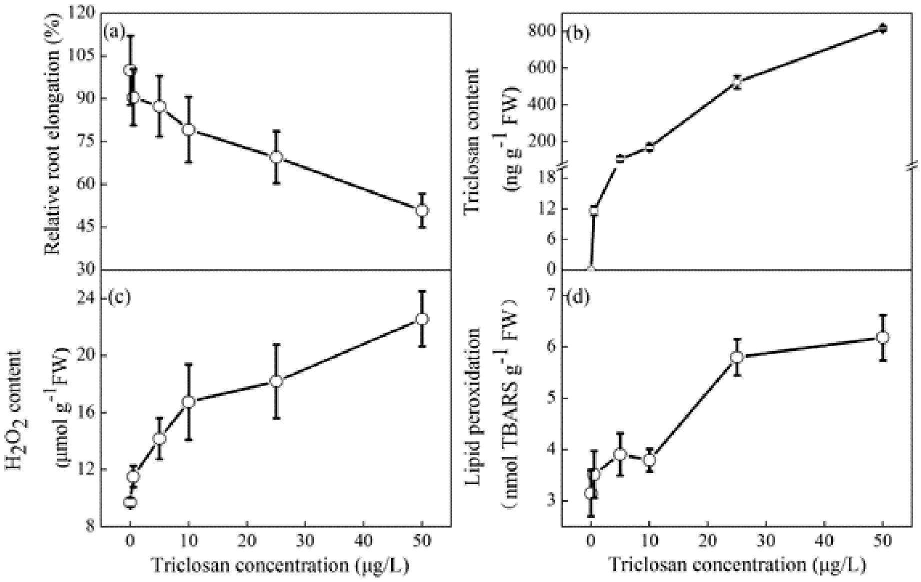


Figure 2. Triclosan dose dependency of (a) root elongation, (b) triclosan accumulation, (c) H_2O_2 content and (d) lipid peroxidation in wheat roots.

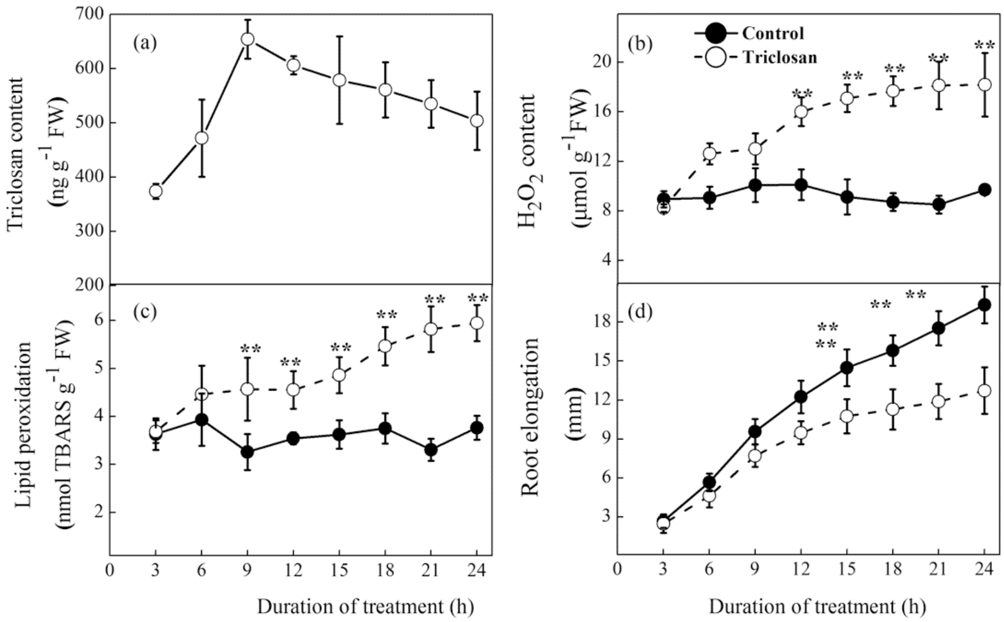
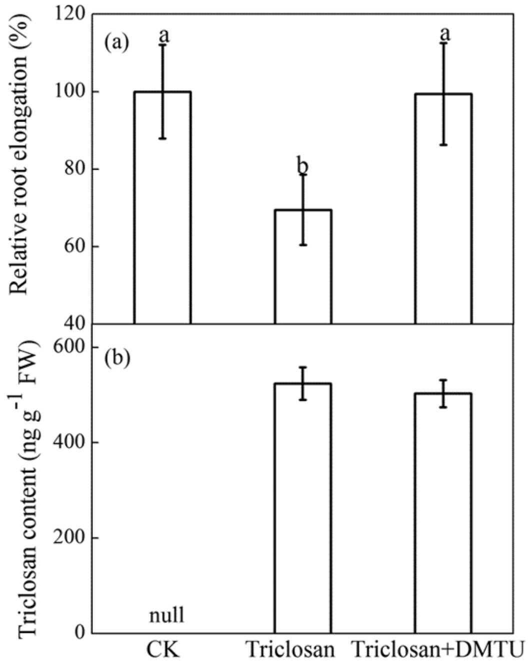


Figure 3. Time course of (a) triclosan accumulation, (b) H_2O_2 content, (c) lipid peroxidation and (d) root elongation of wheat seedlings exposed to triclosan at 25 µg L^{-1}. **, significant difference between triclosan and control treatments at $p < 0.01$.



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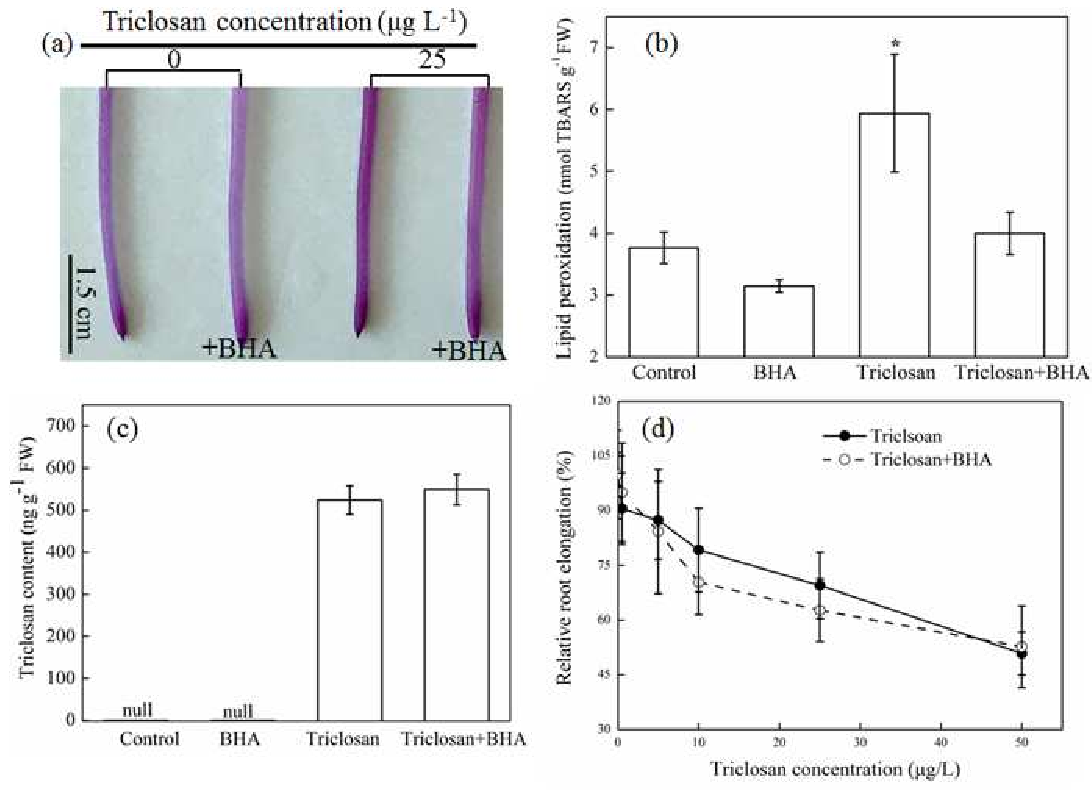

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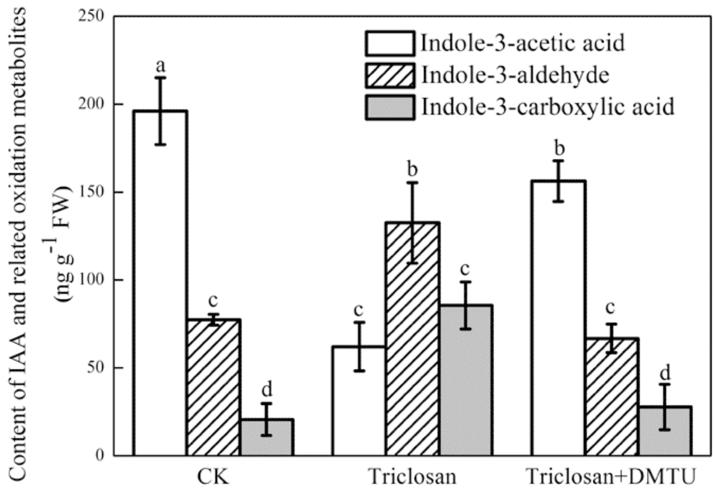

Figure 6. Effects of triclosan and H_2O_2 scavenger (dimethylthiourea, DMTU) on levels of indole-3-acetic acid (IAA) and its two oxidative metabolites (indole-3-aldehyde and indole-3-carboxyloic acid) in wheat roots.

Highlights

1. Triclosan led to a pronounced inhibition of root elongation in wheat seedlings;

2. Eliminating H_2O_2 alleviated triclosan-induced inhibition of root growth;

3. H_2O_2 likely exerted its toxicity by disrupting the auxin signaling pathway.